SiteWatch - Website and Service Monitoring Course:

LINUX and OPERATIONS (1/69) - Linux Automation Project Project: Topic 4, SiteWatch Date prepared: 14 November 2026 Group members: [Add real member names and IDs before submission] ## 1. Objectives Implement a practical Bash automation utility to monitor web URLs and TCP host:port services, record repeated checks, calculate observed uptime, and notify operators only when a monitored target changes state. The tool is meant to run automatically every 3 minutes with a systemd timer. ## 2. System concept 1. Read the external trusted Bash configuration defining URL arrays, service arrays, timeout and success-code range. 2. Validate inputs and confirm necessary commands exist. 3. Check DNS using getent ahosts, TCP using nc -z -w, and HTTP status/response time using curl. 4. Determine UP or DOWN for each target. A URL must pass DNS, TCP, and configured HTTP status checks; a service must pass DNS and TCP. 5. Append a timestamped record for each target to CSV. Load previous state per target and issue alerts only for UP -> DOWN and DOWN -> UP. 6. Atomically update the state file, calculate uptime percentage from CSV, and let systemd repeat the run every 3 minutes. ## 3. Components - bin/sitewatch.sh: main Bash entrypoint, option handling, CSV/state/transition alerts, logging, coordination. - bin/sitewatch-monitor.sh: technical network checks and tab-separated result stream. - bin/sitewatch-report.sh: uptime report from CSV history. - config/sitewatch.conf: monitored targets, timeout, HTTP status limits, data directory. - systemd/sitewatch.service: Type=oneshot invocation of the main script. - systemd/sitewatch.timer: first check after boot and every 3 minutes after a completed check. - install.sh, uninstall.sh: repeatable installation and clean removal. ## 4. Commands and algorithm curl collects the final HTTP code and time_total (seconds), with connection and overall timeouts. getent tests name resolution, and nc checks whether the TCP port is reachable. awk performs numeric processing/reporting, while sed trims user-supplied target whitespace. All major scripts use a shebang and set -euo pipefail; the main script includes main() and additional functions. ## 5. Uptime and files The CSV contains

timestamp,type,target,dns,tcp,http_code,response_sec,status. The state file stores type,target,last_status as TSV. Uptime percentage is 100 * number_of_UP_checks / number_of_all_checks for each target. This is sampled uptime, not continuous time-based availability. All state is keyed by both type and target. ## 6. Configuration and CLI -h shows help; -c FILE selects a configuration; -v displays detailed diagnostic logs; -n validates configuration and lists targets without performing network checks or writing data. Exit codes are: 0 successful operation, 1 configuration/runtime error, and 2 invalid usage. Logs have timestamps and INFO/WARN/ERROR severities; systemd records them in the journal. ## 7. Scheduling, installation and uninstall sudo bash install.sh installs scripts under /usr/local/lib/sitewatch/bin, installs a config to /etc/sitewatch/sitewatch.conf, creates /var/lib/sitewatch, deploys the oneshot service and timer, then enables the timer. Re-running install retains any existing installed config. sudo bash uninstall.sh --purge disables the timer, removes installed files and deletes stored CSV/state. The monotonic timer uses OnBootSec=1min and OnUnitInactiveSec=3min; Persistent=true is only appropriate for supported calendar timers, so it is not used here. ## 8. Validation and demonstration Test cases to capture: 1. Help (-h) and dry-run (-n). 2. A running local Python HTTP server at 127.0.0.1:8000 reports HTTP 200 and UP. 3. Stopping it results in HTTP 000, closed TCP, DOWN and a single alert. 4. Checking again while down creates no duplicate alert. 5. Restarting the web server produces DOWN -> UP recovery log. 6. A non-resolving .invalid domain reports DNS FAIL and DOWN. 7. Missing config gives ERROR/exit 1; invalid option gives exit 2. 8. A ShellCheck pass, timer listing, journal logs, CSV/uptime report. 9. Fresh VM install and complete uninstall. Verification status: Monitoring UP/DOWN and ShellCheck were manually demonstrated on the group’s Ubuntu 26.04 VM; GitHub Actions validates automated branch tests. The course PDF requires Ubuntu Server 22.04/24.04 or Rocky Linux 9, so the final demonstration should be repeated on a compliant VM. No claim is made that the entire system has already been tested on the group’s VM. ## 9. Limitations and next steps The parser handles HTTP/HTTPS URLs with IPv4 or hostnames, but not

bracketed IPv6 hosts. Authentication-protected websites may require different HTTP success criteria. Local demo targets can be changed in the config. The generated PDF is a report draft: add all real student information and the team’s captured screenshots in docs/screenshots/ before submitting. ## 10. Presentation checklist - Install on a clean supported Ubuntu/Rocky VM. - Show systemctl list-timers and journalctl -u sitewatch.service with timestamps. - Run sitewatch.sh -h, -v, and -n by hand. - Simulate a stopped web server and a missing configuration. - Show CSV output, sampled uptime, latest state, and transition alerts. - Run uninstall.sh --purge and confirm removal.